

# ArCo4Archives: A Pattern-based Ontology for Archives

Stella Di Fazio<sup>1,\*†</sup>, Maria Chiara Frangipane<sup>2‡</sup>, Giorgia Lodi<sup>3,\*†</sup>, Cecilia Perrone<sup>3,\*†</sup> and Chiara Veninata<sup>1,†</sup>

<sup>1</sup>Central Institute for Archives - Ministry of Culture, Rome, Italy

<sup>2</sup>BUP srl, Rome, Italy

<sup>3</sup>CNR - Institute of Cognitive Sciences and Technologies, Rome, Italy

## Abstract

This paper introduces ArCo4Archives, a pattern-based ontology developed using the eXtreme Design methodology. It is designed to model the Italian archival domain within the broader existing ArCo ontology network. The ontology, expressed in OWL, addresses the need to represent multilevel archival descriptions, including archival resource sets, units, and items. Additionally, it integrates key concepts for the archival domain, such as identification codes, creators, holders, finding aids, and detailed representations of the institutional profiles of agents involved in archives. The work presented includes formal semantic alignments with international standards like Records in Contexts (RiC-O) and foundational ontologies such as DOLCE Ultra-light (DUL). It also outlines a usage scenario with real-world data and the implementation of SHACL validation rules. These rules are applied when transforming data from the Italian national archive system, managed by the Central Institute for Archives (ICAR) of the Italian Ministry of Culture, into an RDF-based knowledge graph grounded in ArCo4Archives.

## Keywords

Ontology design patterns, Cultural heritage, Archives, Ontologies, Linked Data

## 1. Introduction

The evolution of archival practices has been influenced by the transition from traditional paper-based documentation to digital systems. As archives worldwide undergo digitization, there is a growing need for structured data management systems to ensure the integrity, interoperability, and seamless exchange of information between institutions.

This shift from physical to digital records has not only modernized archival workflows but also introduced new challenges in organizing and accessing vast amounts of cultural heritage data.

While this transition is now widely accepted, it represents a fundamental milestone in archival theory and practice. The demand for consistency and cross-system compatibility has inspired the creation of international archival standards, which provide unified frameworks for organizing, describing, and accessing archival materials across institutional and national boundaries. Specifically, exposing the rich informational archival heritage of national institutions requires ontological models capable of faithfully representing the complex semantics of archival materials while maintaining compatibility with existing administrative workflows and legacy systems.

In this context, the management, dissemination, and long-term preservation of archival cultural heritage also increasingly rely on the Semantic Web paradigm and Linked (Open) Data principles to overcome information silos and achieve cross-domain interoperability. A notable example of this trend is the international reference framework Records in Contexts Ontology (RiC-O) [1], which provides a valuable structural model for the archival sector.

RiC-O represents archival materials in terms of records and provides a general vocabulary and formal rules for describing any kind of archival record resource and its contextual entities. However, archival

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\*Corresponding author.

†These authors contributed equally.

✉ stella.difazio@cultura.gov.it (S. D. Fazio); chiara.frangipane@bupsolutions.com (M. C. Frangipane); giorgia.lodi@cnr.it (G. Lodi); cecilia.perrone@istc.cnr.it (C. Perrone); chiara.veninata@cultura.gov.it (C. Veninata)

ORCID 0000-0001-7116-9338 (G. Lodi)



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description is also deeply rooted in local history and practices that define its specific management. The choice of a model for representing it can sometimes be influenced by the overall technological context in which the description must be integrated and made interoperable. Such a scenario may render a general-purpose model less suitable for the task.

To address these specific needs, we introduce ArCo4Archives, the main contribution of this paper. It is a pattern-based ontology designed to model the Italian archival domain within the broader ArCo ontology network [2], which is used to represent Italian cultural heritage. Developed using the eXtreme Design (XD) methodology [3], ArCo4Archives relies on the reuse of Ontology Design Patterns (ODPs), and it was created, and currently maintained, within a collaboration between the Italian Ministry of Culture, the Institute of Cognitive Sciences and Technologies of CNR and its spinoff to meet the need for representing multilevel archival descriptions, ranging from archival resource sets, units, and items, while preserving semantic alignments with the standards established by the Central Institute for Archives (ICAR) of the same Ministry.

Although ArCo4Archives was designed with the specific characteristics of the Italian archival context in mind, we argue that its use of ontology design patterns may enable its reuse in other local contexts where multilevel descriptions are central to archival data.

Further contributions of this paper include:

- the semantic alignment of ArCo4Archives with international standards like RiC-O and foundational ontologies like DOLCE Ultra Light (DUL);
- a usage example of materialization of ArCo4Archives for real data coming from the Italian National Archival System of ICAR named SIA;
- the work we are conducting on data integrity and quality through the definition of SHACL (Shapes Constraint Language) shapes, created on the basis of specific application requirements derived from the earlier mentioned SIA system.

The rest of this paper is structured as follows. Section 2 discusses relevant work at the state of the art. Section 3 introduces the ArCo4Archives ontology presenting the motivations, the modelling design we adopted and the ontology itself. Section 4 illustrates the ontology semantic alignments to RiC-O and DUL. Section 5 shows how to apply the ontology to real-world data derived from the Italian National Archival System while Section 6 presents the work we are conducting to ensure data quality through the definition of SHACL shapes. Finally, Section 7 concludes the paper and presents some future directions of the work.

## **2. State of the art**

The ontological modeling of the archival domain has been the subject of specific scholarly investigation, addressing concepts such as provenance, agency, and the documentary unit from a semantic web perspective [4]. Some of these works concentrate also on specific archives of cultural relevance [5], where data is released as Linked Open Data using the CIDOC-CRM ontology [6]. Moreover, international and national standards, also developed based on the Semantic Web paradigm, have been defined over time and are the foundations of the work here presented. Thus, in the next subsections we analyze them.

### **2.1. International standards**

At the international level, the standardization of archival description has been coordinated by the International Council on Archives (ICA), which established its first working groups in the 1980s. Building upon the guidelines developed by the International Organization for Standardization (ISO) and through the active contribution of the international archival community, the principal standards for archival description were progressively released: ISAD(G) [7] for the description of archival units, ISAAR(CPF) [8] for archival authority records, ISDF [9] for the description of functions, and ISDIAH [10] for the description of institutions holding archival materials [11].

From the outset, however, it was recognized that these standards would ultimately need to be integrated into a unified conceptual framework. To this end, the Expert Group on Archival Description (EGAD) was established, leading to the development of the Records in Contexts (RiC) project<sup>1</sup>. It includes the four primary standards of archival description, formalized both as a conceptual model (RiC-CM) [12] and an OWL ontology (RiC-O)<sup>2</sup>. A fundamental pillar of the ontology lies in the multidimensional representation of documentary resources, disassociated through the distinction between the Resource and Instantiation classes. Furthermore, the model employs relational classes based on the mechanism of reification, thereby enabling the formalization of complex and dynamic interactions.

Consistent with its foundational design, the ontology does not import directly pre-existing upper-level ontologies; it focuses on the archival domain and establish itself as a self-contained, fully thematic conceptual model.

Alongside the development of conceptual standards, the standardization of archival description also took shape through the definition of XML-based standards for data encoding and exchange, most notably Encoded Archival Description (EAD) [13] and Encoded Archival Context – Corporate Bodies, Persons, and Families (EAC-CPF).

## 2.2. National standards

With regard to Italian initiatives aimed at the formalization of domain ontologies for archival description, several noteworthy projects deserve mention.

In 2012, within the ReLoad project, the Ontology for Archival Description (OAD)<sup>3</sup> was developed to support the publication of archival resources as Linked Data. The ontology incorporates both the EAC-CPF ontology [14], for the description of archival creators, and the Ontology of Cultural Organizations, Services and Access (OCSA) [15].

Between 2014 and 2018, the Central Institute for Archives (ICAR), in collaboration with the MAAS Center (Metodologie e Applicazioni per Archivi Storici), developed the Atlante storico istituzionale dell'Italia unita (ASI)<sup>4</sup>, a linked open data publishing platform mapping the evolution of the Italian administrative districts from 1861 to 2013, which was accompanied by the creation of a dedicated ontology bearing the same name [16].

The SAN ontology<sup>5</sup>, developed by the Central Institute for Archives (Istituto Centrale per gli Archivi, ICAR) in 2013, represents the initiative most closely related to the work presented in this paper. It provides an OWL representation of the CAT-SAN exchange schema, an XML-based data exchange format adopted by the National Archival System (SAN) [17], which at that time was the national aggregator and access point for searching and consulting Italy's archival heritage.

The SAN ontology however, is not integrated in the general cultural heritage semantic framework grounded in ArCo [2] that in the meantime was developed in Italy.

The ArCo4Archives ontology presented in this paper thus born for extending the ArCo ontology network by introducing a dedicated ontology module for archival description.

ArCo is an ontology network developed by the Italian Central Institute for Catalogue and Documentation (ICCD) of the Italian Ministry of Culture (MIC) in collaboration with the Institute of Cognitive Sciences and Technologies (ISTC) of the National Research Council of Italy (CNR), and published between 2018 and 2019 to support knowledge representation in the cultural heritage domain. ArCo [2], [18] was designed to shift the representation from the cataloguing records developed by ICCD to the cultural properties themselves, which become the nucleus of the representation. The ontology network is organized into several modules to cover context descriptions, cultural events, locations, specific vertical elements like clothing and so on.

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<sup>1</sup><https://ica-egad.github.io/RiC-O/about.html>

<sup>2</sup><https://www.ica.org/standards/RiC/ontology>

<sup>3</sup><http://lod.xdams.org/reload/oad>

<sup>4</sup><http://dati.san.beniculturali.it/asi/local/>

<sup>5</sup><http://www.san.beniculturali.it/SAN/san-lod.owl>

In general, ArCo is formally semantically aligned with the network of ontologies for the Italian Public Sector<sup>6</sup>, and provides alignment relationships to widely adopted standards and ontologies of the domain like CIDOC CRM [6] and Europeana Data Model (EDM) [19].

### 3. Ontology overview

This section aims to explain the rationale behind the creation of the ontology, the process we followed for its development and the key decisions taken regarding the modeling of archival information.

#### 3.1. Motivations

The choice to develop a vertical and domain-specific ontological model for the archival sector, rather than the immediate adoption of broad-spectrum reference frameworks such as RiC-O<sup>7</sup> or CIDOC-CRM[6], is motivated by three fundamental aspects.

- *Publication of ICAR's information heritage using Semantic Web standards*: there is a recognized priority need to expose the information heritage pertaining to the Archival Information System (SIA) of the Central Institute for Archives (ICAR) using Semantic Web standards for integration with already available data about other cultural properties. The consolidated adoption within ICAR of XML-based standards for the representation of archival data makes it appropriate to define an ontology that is formally co-extensive and structurally adherent to these standards, in order to mitigate semantic asymmetries and optimize the processes of transformation, alignment, and homogenization of data.
- *Representation of national taxonomic and archival-system specificities*: archival descriptive practices are intrinsically connected to national legal, administrative, and historiographical systems. Ontological models of general or supranational scope frequently reveal an insufficient level of granularity to capture the structural specificities, chronological designations, and typological hierarchies proper to the Italian archival tradition. The development of a specific model ensures the preservation of semantic fidelity to the sources while respecting domestic descriptive practices.
- *Semantic integration within the cultural heritage ecosystem*: in Italy, ArCo become a consolidated national standard to describe cultural properties in general. Integrating in it archival descriptions allows for data integration for different types of cultural heritage properties (archives, libraries, museums). This also helps enhance semantic interoperability with major digitization initiatives in the Italian cultural sector, also including the IPAC Italian data space of the Digital Library Institute of the Ministry of Culture<sup>8</sup>. In general, the harmonization of archival descriptors with cross-domain already existing standards fosters the overcoming of information silos, enabling a unified and transversal access model to heterogeneous cultural resources.

#### 3.2. The ArCo4Archives ontology

Remaining faithful to the archival information representation model established by the ICA standards (General International Standard Archival Description), the proposed modeling approach that led to the definition of ArCo4Archives is rooted in the idea that an archival resource is a cultural property. This shift aligns with the transition observed in ArCo, moving from a focus on the record to the cultural asset itself.

In general, the modeling methodology used to create ArCo4Archives is eXtreme Design, an agile, incremental and iterative method for ontology design[3]. It is worth noticing that since ArCo4Archives originates from the requirement to incorporate archival domain data within the consolidated framework of the ArCo ontology network[2], many design principles of ArCo4Archives derive also from ArCo.

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<sup>6</sup><https://schema.gov.it>

<sup>7</sup><https://www.ica.org/standards/RiC/ontology>

<sup>8</sup><https://ecomonic.cultura.gov.it/ipac/>

The methodology foresees a first phase of requirement collection. To this end, a set of competency questions specifically targeting archival information were formulated together with the domain experts of the Italian Ministry of Culture. The competency questions served to guide the modeling process, and a small excerpt of them is presented in Table 1.

**Table 1**  
Competency questions elicited from the scenario.

| ID  | CQ                                                                                                                                         |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------|
| CQ1 | Which archival resources are restricted from public access, and why (condition of access type)?                                            |
| CQ2 | Which archival resources are documented/described by a given documentation relation (internal or external to the SIA system)?              |
| CQ3 | Which archival resources have a copy/original relation with a given archival resource (i.e., which is the original and which is the copy)? |
| CQ4 | How many and which archival fonds document assistance and charitable activities in a specific city or province?                            |
| CQ5 | What is the numbering system applied to a given archival unit or item?                                                                     |
| CQ6 | Which is the total extent of archival resource X?                                                                                          |
| CQ7 | Which is the archival reference code of the archival resource X?                                                                           |
| CQ8 | What is the level of description (fonds, series, unit, item, etc.) of a given archival resource?                                           |

Additionally, ArCo4Archives was developed by reusing Ontology Design Patterns (ODP)[20]. Relying on the ArCo ontology network backbone, ArCo4Archives extends certain established patterns used in ArCo that allow for a better description of the archival data.

Table 2 lists the main ontology design patterns that have been used from ArCo and materialised/specialised in the ontology.

**Table 2**  
Ontology Design Patterns used in ArCo4Archives

| ODP                  | Description wrt ArCo4Archives                                                                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Classification       | Used to classify all the different types of archival resources                                                                                                                     |
| Collection Entity    | Used in ArCo and specialised in the domain to represent the archival resources that are collections                                                                                |
| TimeIndexedSituation | Used in ArCo and directly used for the Location and the Designation of the archival resource. Also specialized to represent the various responsibilities on the archival resources |
| TimeInterval         | Through the use of national ontologies about time for the public sector where a temporal entity is also a time interval                                                            |
| Description          | Specialised for specific domain classes like Arrangement                                                                                                                           |

These ontology design patterns are widely adopted across various domains and can be particularly beneficial in the archival sector. Archival data is inherently structured around robust classifications of different document types and clearly defined collections of documents. As such, Classification and Collection Entity ODPs are relevant, as they provide a structured way to represent hierarchical relationships and groupings of archival materials. Furthermore, the archival domain places emphasis on temporal aspects, particularly the ability to track changes over time. To address this, the Time Interval and Time-Indexed Situation patterns are reused. The Time Interval pattern allows for the representation of temporal boundaries, enabling the modeling of events or states that occur within specific timeframes. Meanwhile, the Time-Indexed Situation pattern facilitates the description of entities or conditions that evolve over time, ensuring that historical context and temporal dynamics are captured. By leveraging these patterns, archival ontologies can achieve greater interoperability, and alignment with both domain-specific and cross-domain standards. This not only enhances the semantic richness of the data but also supports potential more sophisticated querying and reasoning capabilities, which can be useful for historical research and archival management.

Finally, another important modelling choice that was made is to align ArCo4Archives to an upper-level ontology developed in the national context for the public sector named *l0*<sup>9</sup>. This upper ontology is in turn fully aligned with the DOLCE Ultra-Light (DUL) foundational ontology, thus allowing us to use DOLCE as well when modeling archival data.

Based on these modeling principles, Figure 1 illustrates a simplified graphical version of the ontology,

<sup>9</sup><https://w3id.org/italia/onto/l0>





It models the historical and administrative relationships between institutions and various contexts, such as successions and hierarchical dependencies, while also capturing the relationship with the relevant territory through the `clv:hasSpatialCoverage` property.

## 4. Semantic alignments with existing ontologies

To preserve a high degree of interoperability of the system, an alignment process was conducted between the entities of the ArCo4Archives ontology and those of Records in Contexts Ontology (RiC-O), recognized as the international standard for ontological modeling in the archival domain, and DUL foundational ontology.

This semantic alignment activity uncovered compelling insights that, in several instances, corroborated the necessity of developing a dedicated ontological module tailored to the Italian context[23].

Some of these alignments are listed in Table3. As a preliminary methodological note, semantic alignments were predominantly defined using OWL (Ontology Web Language) constructs, specifically by establishing ArCo4Archives entities as specializations of RiC-O entities, except where a precise, direct equivalence was readily identifiable. Certain alignment mappings have been formalized in the RDF/Turtle serialization and are currently published in the official ICAR GitHub repository<sup>13</sup>, subject to upcoming revisions and extensions.

**Table 3**

Alignments with other ontologies.  $\sqsubseteq$  denotes subclass or subproperty alignment.  $\equiv$  denotes equivalent alignment

| ArCo element              | Alignment     | Other ontologies elements                          |
|---------------------------|---------------|----------------------------------------------------|
| a-arc:ArchivalResource    | $\sqsubseteq$ | rico:RecordResource and dul:Entity                 |
| a-arc:ArchivalResourceSet | $\sqsubseteq$ | rico:RecordSet and dul:Collection                  |
| a-arc:ArchivalUnit        | $\sqsubseteq$ | rico:RecordSet and dul:Collection                  |
| a-arc:ArchivalItem        | $\sqsubseteq$ | rico:Record and dul:Entity                         |
| a-arc:Event               | $\equiv$      | rico:Event                                         |
| a-arc:totalExtentInstring | $\equiv$      | rico:recordResourceExtent                          |
| a-arc:Arrangement         | $\sqsubseteq$ | dul:Description (and <code>l0:Description</code> ) |
| a-arc:Extent              | $\sqsubseteq$ | dul:Object (and <code>l0:Object</code> )           |

In general, the mapping process has highlighted some structural differences between ArCo4Archives and RiC-O.

As previously described, the foundational premise of ArCo4Archives is the representation of the three principal levels of archival description; namely, the archival resource set, the archival unit, and the archival item, adhering to the principle of multilevel description. In contrast, RiC-O favors an approach centered on individual records, represented by the `rico:Record` class and its potential aggregations (`rico:RecordSet`).

Finally, owing to its adoption of Ontology Design Patterns (ODPs), the ArCo4Archives modeling framework exhibits a higher degree of granularity and a better possibility to align with already existing ontologies of the Web of Data. This structural granularity does not allow a direct 1-to-1 entity alignment, thereby necessitating the use of more sophisticated axioms, currently undergoing further evaluation.

## 5. Usage scenario for the Italian National Archive System

The following RDF triples, serialized in RDF/turtle (for the sake of conciseness we do not include prefixes), provides a small usage example of the ArCo4Archives materialization. The data is provided by ICAR through a so-called ICAR-Import 2.0 XML example of the description of an archival resource

<sup>13</sup><https://github.com/IstitutoCentraleArchivi-ICAR/ICAR-Semantics/tree/main>

set.<sup>14</sup> ICAR Import 2.0 is the official XML standard used for data exchange and interoperability with the SIA system<sup>15</sup>. It aligns with the METS ECO-MiC format<sup>16</sup> for managing digital objects.

```
:archival-resource-set/IT-ASBZ-F230000094 a a-arc:ArchivalResourceSet ;
  rdfs:label "Intendenza di finanza di Bolzano - IT-ASBZ-F230000094";
  a-arc:hasArrangement :arrangement/IT-ASBZ-F230000094 ;
  a-arc:hasExtent :extent/84e8eb01913893ddca6248fc6ae10e6e ;
  a-arc:isDescribedBy :finding-aid/6782bbfdf5e31a163b568b3ed4cff81e ;
  a-arc:totalExtentInString "300 m.l. (10 volumi e 400 buste)"^^xsd:string;
  a-arc:totalExtentInLinearMeters "300"^^xsd:integer;
  a-lite:alternativeDesignation "Delegazione di finanza di Bolzano"^^xsd:string;
  a-cd:hasSubject :subject/guerra , :subject/storia ;
  a-lite:currentDesignation "Intendenza di finanza di Bolzano"^^xsd:string;
  ti:atTime :time-interval/1941-1984 ;
  a-lite:hasArchivalCreator :agent/IT-ASBZ-CC236400094 .
```

The RDF excerpt represents the particular archival resource set called "Intendenza di finanza di Bolzano" with its main elements. The extent of this particular archival complex is measured in linear meters ("300 l.m."), counting 10 volumes and 400 envelopes. Furthermore, the example shows its current designation, the time interval it refers to, and the creator. In this latter case, it is worth noticing the shortcut defined in the modeling we use without materializing the complex pattern earlier described about the creator responsibility. This shortcut is useful when not all the data about responsibility are available. The archival resource set is also described by the following finding aid:

```
:finding-aid/6782bbfdf5e31a163b568b3ed4cff81e a a-arc:FindingAid ;
  rdfs:label "Fonti per la storia dell'Alto Adige
  e del Sud Tirolo, Bolzano, 1998.".
```

## 6. Data quality validation with SHACL Shape

To ensure the integrity, consistency, and overall reliability of the archival knowledge graph, a validation layer is currently under implementation using SHACL (Shapes Constraint Language)<sup>17</sup>.

In this architectural context, the SHACL shapes act as formal quality-assurance contracts. Rather than being defined arbitrarily, these constraints were derived from custom templates and domain specifications provided by ICAR. These specifications operationalize the stringent logical constraints, cardinalities (such as minimum and maximum occurrence of properties), and mandatory requirements natively enforced by the legacy Archival Information System (SIA).

Below we report a small example of SHACL rules written for managing data on the extent of an archival resource set. Other SHACL shapes can be found in the related ICAR's Github repository<sup>18</sup>. In the SIA system, the extent information is mandatory, and its value must be specified as an integer.

```
ex:ArchivalResourceSetShape
  a sh:NodeShape ;
  sh:targetClass a-arc:ArchivalResourceSet ;
  sh:property [
    sh:path a-arc:hasExtent ;
    sh:severity sh:Violation ;
    sh:class a-arc:Extent ;
```

<sup>14</sup>[https://github.com/IstitutoCentraleArchivi-ICAR/Esempi-ICAR-import-2/blob/main/Tracciati\\_EAD3/ComplArch\\_SIA.xml](https://github.com/IstitutoCentraleArchivi-ICAR/Esempi-ICAR-import-2/blob/main/Tracciati_EAD3/ComplArch_SIA.xml)

<sup>15</sup><https://icar.cultura.gov.it/attivita-e-progetti/progetti-icar-1/icar-import-2>

<sup>16</sup><https://ecommic.cultura.gov.it/ipac/2024/04/19/mets-eco-mic/>

<sup>17</sup><https://www.w3.org/TR/shacl/>

<sup>18</sup><https://github.com/IstitutoCentraleArchivi-ICAR/ICAR-Semantics/tree/main/assets>

```

sh:message "The relationship between the archival resource set and its extent
  is mandatory: the property a-arc:hasExtent must be materialized. "
] ,
[
  sh:path a-arc:totalExtentInLinearMeters ;
  sh:datatype xsd:integer ;
  sh:severity sh:Violation ;
  sh:message "The relationship between the extent and its type is mandatory:
    a-arc:totalExtentInLinearMeters must be materialized."
] .

```

In essence, this approach ensures that the resulting Linked Data not only complies with semantic web standards, but also faithfully mirrors the structural and administrative integrity required by the Italian national archival standards.

## 7. Conclusions and future work

In this paper, we presented ArCo4Archives, an ontology developed in collaboration with the Central Institute for Archives of the Italian Ministry of Culture. This ontology is designed to represent archival data as collected in the SIA System, capturing the unique characteristics of archival practices in Italy, although we claim the model can be reused also in other local contexts with similar characteristics. It offers a more specialised modeling approach compared to international standards such as RiC-O, while ensuring semantic alignment with these standards. To guarantee data quality, a set of SHACL shapes has been defined. Examples of these shapes, along with real data from the Italian National Archival System, are provided in the paper.

Future work will focus on extending and consolidating certain elements of the presented ontology that is available in its first version, and materializing in the data the RiC-O alignments so that to allow users to query the data according to their preferred standard. Additionally, we plan to carry out an in depth evaluation with different archivists, and publish the ontology in schema.gov.it, the national catalog of semantic assets. Since the results presented in the paper are also considered in the context of the European project, INFINITY<sup>19</sup> we plan to assess their applicability in possible different local contexts.

Finally, ArCo4Archives will also serve as the foundational layer for developing the new Italian National Archival System, which will automatically produce Linked Data grounded in ArCo4Archives. This Linked Data may also be integrated to the broader Italian cultural heritage domain, already available on the Web as an RDF-based knowledge graph.

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## Declaration on Generative AI

In the preparation of this work, the authors used Gemini and Mistral in order to apply grammar and spelling checks. After using these tools, the authors reviewed and edited the content as needed and assume full responsibility for the content of the publication.

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<sup>19</sup><https://infinity-eccch.eu/>

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